Chapter 1, Problem 10PP

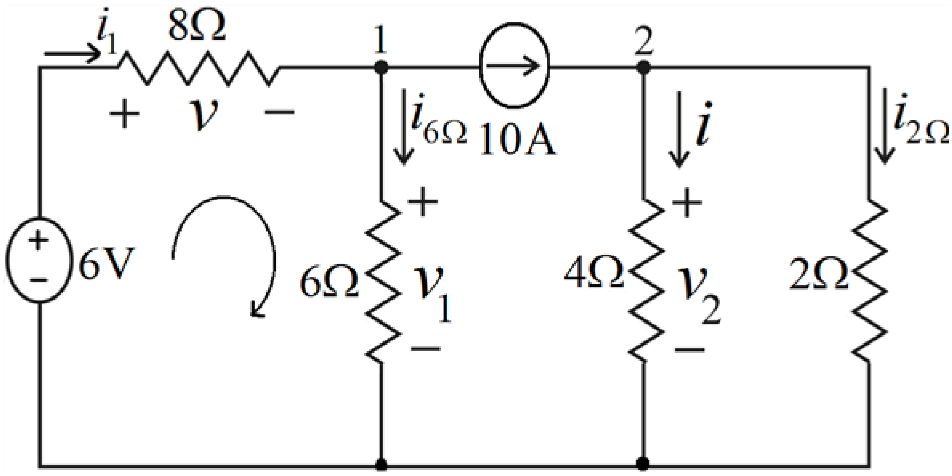
Problem

Try applying this process to some of the more difficult problems at the end of the chapter.

Step-by-step solution

Step 1 of 8

Consider the circuit as shown below:



Step 2 of 8

Let us the find the values of v and i.

To determine the values of v and i, we use Ohm's law and Kirchhoff's laws.

From Ohm's law,

$$v = i_1 \times 8$$

$$v = 8i_1 \quad \text{.............. (1)}$$

Current through resistor 6Ω is,

$$v_1 = i_{6\Omega} \times 6$$

$$i_{6\Omega} = \frac{v_1}{6} \quad \text{.............. (2)}$$

Step 3 of 8

Current through resistor 4Ω is,

$$v_2 = i \times 4$$

$$i = \frac{v_2}{4} \quad \ldots\ldots\ldots\ldots. (3)$$

Since the 4Ω and 2Ω resistors are in parallel, voltage across them is same.

Current through resistor 4Ω is,

$$v_2 = i_{2\Omega} \times 2$$

$$i_{2\Omega} = \frac{v_2}{2} \quad \ldots\ldots\ldots\ldots. (4)$$

Step 4 of 8

Applying Kirchhoff's voltage law to the left loop, we obtain

$$-6 + v + v_1 = 0$$

$$v_1 = 6 - v \quad \ldots\ldots\ldots\ldots. (5)$$

Apply Kirchhoff's current law at node 1:

$$i_1 - i_{6\Omega} - 10 = 0$$

Step 5 of 8

Substituting i_1 from equation (1), $i_{6\Omega}$ from equation (2) into above equation gives,

$$\frac{v}{8} - \frac{v_1}{6} - 10 = 0$$

$$\frac{3v - 4v_1}{24} = 10$$

$$3v - 4v_1 = 240$$

Step 6 of 8

Substituting v_1 from equation (5) into above equation gives,

$$3v - 4(6 - v) = 240$$

$$7v = 216$$

$$v = \frac{216}{7}$$

$$\therefore \boxed{v = 30.86\text{ V}}$$

Step 7 of 8

Apply Kirchhoff's current law at node 2:

$$10 - i - i_{2\Omega} = 0$$

Substituting i from equation (3), $i_{2\Omega}$ from equation (4) into above equation gives,

$$10 - \frac{v_2}{4} - \frac{v_2}{2} = 0$$

$$10 = \frac{v_2 + 2v_2}{4}$$

$$\frac{3v_2}{4} = 10$$

Step 8 of 8

$$v_2 = \frac{40}{3}\text{ V}$$

Substituting v_2 value in equation (3), we obtain

$$i = \frac{\left(\frac{40}{3}\right)}{4}$$

$$i = \frac{10}{3}$$

$$\therefore \boxed{i = 3.33\text{ A}}$$